

Chapter 25

Probability

Class - 8 Concise Mathematics Selina



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Exercise 25

Question 1(i)

A coin is tossed three times, the number of possible outcomes are :

1. 8
2. 6
3. 10
4. 4



Answer

When a coin is tossed, each toss has 2 possible outcomes : head (H) or tail (T).

The number of possible outcomes when a coin is tossed n times = 2^n

For this case, $n = 3$

$$\text{So, } 2^3 = 8$$

The number of possible outcomes when a coin is tossed 3 times is 8.

Hence, option 1 is the correct option.

Question 1(ii)

If $P(A)$ denotes the probability of getting an event A . Then $P(\text{not getting } A)$ is :

1. $1 - P(A)$

2. $P(A) - 1$

3. $P(A)$

4. $1 + P(A)$

Answer

The probability of not getting an event is the complement of the probability of getting the event, which is,

$$P(\text{not getting } A) = 1 - P(A)$$

Hence, option 1 is the correct option.

Question 1(iii)

A coin is tossed once. The probability of getting a tail is :

1. 1
2. 2
3. $\frac{1}{2}$
4. none of these

Answer

When a coin is tossed, each toss has 2 possible outcomes : head (H) or tail (T).

The number of possible outcomes when a coin is tossed n times = 2^n

For this case, n = 1

So, $2^1 = 2$

The number of possible outcomes when a coin is tossed once is 2.

Number of favourable outcomes (Getting tail (T)) = 1

$$\begin{aligned} P(\text{Getting tail (T)}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{1}{2} \end{aligned}$$

Hence, option 3 is the correct option.

Question 1(iv)

A coin is tossed two times. The probability of getting atleast one tail is :

1. $\frac{3}{4}$

2. $\frac{4}{3}$

3. $\frac{1}{2}$

4. $\frac{5}{6}$

Answer

When a coin is tossed, each toss has 2 possible outcomes : head (H) or tail (T).

The number of possible outcomes when a coin is tossed n times = 2^n

For this case, n = 2

So, $2^2 = 4$

The number of possible outcomes when a coin is tossed 2 times is 4.

Number of favourable outcomes (Getting at least one tail) = 3 (HT, TH, TT)

$$P(\text{Getting at least one tail}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{3}{4}$$

Hence, option 1 is the correct option.

Question 1(v)

A card is drawn from a well shuffled pack of 52 playing cards. The probability of getting a face card is :

1. $\frac{4}{13}$

2. $\frac{2}{13}$

3. $\frac{3}{13}$

4. $\frac{1}{13}$

Answer

Total number of outcomes = 52

Number of favourable outcomes (Getting a face card) = 12

$$P(\text{Getting a face card}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{12}{52}$$

$$= \frac{3}{13}$$

Hence, option 3 is the correct option.

Question 2

A coin is tossed twice. Find the probability of getting :

(i) exactly one head

(ii) exactly one tail

(iii) two tails

(iv) two heads

Answer

When a coin is tossed, each toss has 2 possible outcomes : head (H) or tail (T).

The number of possible outcomes when a coin is tossed n times = 2^n

For this case, $n = 2$

$$\text{So, } 2^2 = 4$$

The number of possible outcomes when a coin is tossed 2 times is 4.

(i) Number of favourable outcomes (Getting exactly one head) = 2 (HT, TH)

Total number of outcomes = 4

$$\begin{aligned} P(\text{Getting exactly one head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Hence, the probability of getting exactly one head is $\frac{1}{2}$.

(ii) Number of favourable outcomes (Getting exactly one tail) = 2 (HT, TH)

Total number of outcomes = 4

$$\begin{aligned}P(\text{Getting exactly one tail}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{2}{4} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting exactly one tail is $\frac{1}{2}$.

(iii) Number of favourable outcomes (Getting two tails) = 1 (TT)

Total number of outcomes = 4

$$\begin{aligned}P(\text{Getting two tails}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{1}{4}\end{aligned}$$

Hence, the probability of getting two tails is $\frac{1}{4}$.

(iv) Number of favourable outcomes (Getting two heads) = 1 (HH)

Total number of outcomes = 4

$$\begin{aligned}P(\text{Getting two heads}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{1}{4}\end{aligned}$$

Hence, the probability of getting two heads is $\frac{1}{4}$.

Question 3

A letter is chosen from the word 'PENCIL', what is the probability that the letter chosen is a consonant ?

Answer

Total number of outcomes = 6

Number of favourable outcomes (Getting a consonant) = 4 (P, N, C, L)

$$\begin{aligned} P(\text{Getting a consonant}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{4}{6} \\ &= \frac{2}{3} \end{aligned}$$

Hence, the probability of getting a consonant is $\frac{2}{3}$.

Question 4

A bag contains a black ball, a red ball and a green ball, all the balls are identical in shape and size. A ball is drawn from the bag without looking into it. What is the probability that the ball drawn is :

- (i) a red ball
- (ii) not a red ball
- (iii) a white ball.

Answer

(i) Total number of outcomes = Total number of balls = 1 black ball + 1 red ball + 1 green ball = 3

Number of favourable outcomes (Getting a red ball) = 1

$$P(\text{Getting a red ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$
$$= \frac{1}{3}$$

Hence, the probability of getting a red ball is $\frac{1}{3}$.

(ii) Total number of outcomes = Total number of balls = 1 black ball + 1 red ball + 1 green ball = 3

Number of favourable outcomes (Getting not a red ball) = 2 (black ball, green ball)

$$P(\text{Getting not a red ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$
$$= \frac{2}{3}$$

Hence, the probability of getting not a red ball is $\frac{2}{3}$.

(iii) Total number of outcomes = Total number of balls = 1 black ball + 1 red ball + 1 green ball = 3

Number of favourable outcomes (Getting a white ball) = 0

$$P(\text{Getting a white ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{0}{3}$$
$$= 0$$

Hence, the probability of getting a white ball is 0.

Question 5

In a single throw of a die, find the probability of getting a number

- (i) greater than 2
- (ii) less than or equal to 2
- (iii) not greater than 2.

Answer

(i) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting a number greater than 2) = 4 (3, 4, 5 and 6)

$$P(\text{Getting a number greater than 2}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$
$$= \frac{4}{6}$$
$$= \frac{2}{3}$$

Hence, the probability of getting a number greater than 2 is $\frac{2}{3}$.

(ii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting a number less than or equal to 2) = 2
(1, 2)

$P(\text{Getting a number less than or equal to 2}) =$

$\frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

$= \frac{2}{6}$

$$= \frac{1}{3}$$

Hence, the probability of getting a number less than or equal to 2 is $\frac{1}{3}$.

(iii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting a number not greater than 2) = 2 (1, 2)

$P(\text{Getting a number not greater than 2}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

$$= \frac{2}{6}$$

$$= \frac{1}{3}$$

Hence, the probability of getting a number not greater than 2 is $\frac{1}{3}$.

Question 6

A bag contains 3 white, 5 black and 2 red balls, all of the same size. A ball is drawn from the bag without looking into it, find the probability that the ball drawn is :

(i) a black ball

(ii) a red ball

(iii) a white ball

(iv) not a red ball

(v) not a black ball

Answer

(i) Total number of possible outcomes = Total number of balls = 3 white balls + 5 black balls + 2 red balls = 10

Number of favourable outcomes (Getting a black ball) = 5

$$\begin{aligned}P(\text{Getting a black ball}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{5}{10} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting a black ball is $\frac{1}{2}$.

(ii) Total number of possible outcomes = Total number of balls = 3 white balls + 5 black balls + 2 red balls = 10

Number of favourable outcomes (Getting a red ball) = 2

$$P(\text{Getting a red ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{2}{10}$$
$$= \frac{1}{5}$$

Hence, the probability of getting a red ball is $\frac{1}{5}$.

(iii) Total number of possible outcomes = Total number of balls = 3 white balls + 5 black balls + 2 red balls = 10

Number of favourable outcomes (Getting a white ball) = 3

$$P(\text{Getting a white ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$
$$= \frac{3}{10}$$

Hence, the probability of getting a white ball is $\frac{3}{10}$.

(iv) Total number of possible outcomes = Total number of balls = 3 white balls + 5 black balls + 2 red balls = 10

Number of favourable outcomes (Getting not a red ball) = Number of white balls + Number of black balls = 3 + 5 = 8

$$P(\text{Getting not a red ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$
$$= \frac{8}{10}$$
$$= \frac{4}{5}$$

Hence, the probability of getting not a red ball is $\frac{4}{5}$.

(v) Total number of possible outcomes = Total number of balls = 3 white balls + 5 black balls + 2 red balls = 10

Number of favourable outcomes (Getting not a black ball) = Number of white balls + Number of red balls = 3 + 2 = 5

$$\begin{aligned} P(\text{Getting not a black ball}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{5}{10} \\ &= \frac{1}{2} \end{aligned}$$

Hence, the probability of getting not a black ball is $\frac{1}{2}$.

Question 7

In a single throw of a die, find the probability that the number :

- (i) will be an even number
- (ii) will be an odd number
- (iii) will not be an even number.

Answer

(i) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting an even number) = 3 (2, 4, 6)

$$\begin{aligned}P(\text{Getting an even number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{3}{6} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting an even number is $\frac{1}{2}$.

(ii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting an odd number) = 3 (1, 3, 5)

$$\begin{aligned}P(\text{Getting an odd number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{3}{6} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting an odd number is $\frac{1}{2}$.

(iii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting not an even number) = 3 (1, 3, 5)

$$\begin{aligned}P(\text{Getting not an even number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{3}{6} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting not an even number is $\frac{1}{2}$.

Question 8

In a single throw of a die, find the probability of getting :

- (i) 8
- (ii) a number greater than 8
- (iii) a number less than 8

Answer

(i) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting 8) = 0

$$\begin{aligned} P(\text{Getting 8}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{0}{6} \\ &= 0 \end{aligned}$$

Hence, the probability of getting 8 is 0.

(ii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcome (Getting a number greater than 8) = 0

$$\begin{aligned} P(\text{Getting a number greater than 8}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{0}{6} \end{aligned}$$

$$= 0$$

Hence, the probability of getting a number greater than 8 is 0.

(iii) Total number of possible outcomes = 6 (1, 2, 3, 4, 5 and 6)

Number of favourable outcomes (Getting a number less than 8) = 6 (1, 2, 3, 4, 5 and 6)

$$P(\text{Getting a number less than 8}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{6}{6}$$

$$= 1$$

Hence, the probability of getting a number less than 8 is 1.

Question 9

Which of the following cannot be the probability of an event ?

(i) $\frac{2}{7}$

(ii) 3.8

(iii) 127 %

(iv) -0.8

Answer

The probability of an event always lies between 0 and 1. Therefore, any value less than 0 or greater than 1 cannot be the probability of an event.

(i) $\frac{2}{7}$ is between 0 and 1, so it can be a probability.

(ii) 3.8 is greater than 1, so it cannot be a probability.

(iii) 127% is equivalent to 1.27, which is greater than 1, so it cannot be a probability.

(iv) -0.8 is less than 0, so it cannot be a probability.

Question 10

A bag contains six identical black balls. A boy withdraws one ball from the bag without looking into it. What is the probability that he takes out :

(i) a white ball ?

(ii) a black ball ?

Answer

(i) Total number of possible outcomes = 6 (six black balls)

Number of favourable outcomes (Getting a white ball) = 0

$$P(\text{Getting a white ball}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

$$= \frac{0}{6}$$

$$= 0$$

Hence, the probability of getting a white ball is 0.

(ii) Total number of possible outcomes = 6 (six black balls)

Number of favourable outcomes (Getting a black ball) = 6

$$\begin{aligned}P(\text{Getting a black ball}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{6}{6} \\&= 1\end{aligned}$$

Hence, the probability of getting a black ball is 1.

Question 11

Three identical coins are tossed together. What is the probability of obtaining :

- (i) all heads ?
- (ii) exactly two heads ?
- (iii) exactly one head ?
- (iv) no head ?

Answer

When three identical coins are tossed together, the total number of possible outcomes = 8 (i.e. HHH, HHT, HTH, THH, TTH, THT, HTT and TTT)

- (i) Number of favourable outcomes (Getting all heads) = 1 (HHH)

$$\begin{aligned}P(\text{Getting all heads}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\&= \frac{1}{8}\end{aligned}$$

Hence, the probability of getting all heads is $\frac{1}{8}$.

(ii) Number of favourable outcomes (Getting exactly two heads) = 3 (HHT, HTH, THH)

$$\begin{aligned} P(\text{Getting exactly two heads}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{3}{8} \end{aligned}$$

Hence, the probability of getting exactly two heads is $\frac{3}{8}$.

(iii) Number of favourable outcomes (Getting exactly one head) = 3 (HTT, TTH, THT)

$$\begin{aligned} P(\text{Getting exactly one head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{3}{8} \end{aligned}$$

Hence, the probability of getting exactly one head is $\frac{3}{8}$.

(iv) Number of favourable outcomes (Getting no head) = 1 (TTT)

$$\begin{aligned} P(\text{Getting no head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{1}{8} \end{aligned}$$

Hence, the probability of getting exactly no head is $\frac{1}{8}$.

Question 12

A book contains 92 pages. A page is chosen at random. What is the probability that the sum of the digits in the page number is 9 ?

Answer

Total number of possible outcomes = 92

Number of favourable outcomes (Getting a page where the sum of the digits in the page number is 9) = 10 (9, 18, 27, 36, 45, 54, 63, 72, 81, 90)

$P(\text{Getting a page where the sum of the digits in the page number is 9}) =$

$\frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$

$= \frac{10}{92}$

$$= \frac{5}{46}$$

Hence, the probability of getting a page where the sum of the digits in the page number is 9 is $\frac{5}{46}$.

Question 13

Two coins are tossed together. What is the probability of getting :

(i) at least one head ?

(ii) both heads or both tails ?

Answer

When two coins are tossed together, the total number of possible outcomes = 4
(i.e. HH, HT, TH and TT)

(i) Number of favourable outcomes (Getting at least one head) = 3 (HT, TH and HH)

$$\begin{aligned} P(\text{Getting at least one head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{3}{4} \end{aligned}$$

Hence, the probability of getting at least one head is $\frac{3}{4}$.

(ii) Number of favourable outcomes (Getting both heads or both tails) = 2 (HH and TT)

$$\begin{aligned} P(\text{Getting both heads or both tails}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{2}{4} \\ &= \frac{1}{2} \end{aligned}$$

Hence, the probability of getting both heads or both tails is $\frac{1}{2}$.

Question 14

From 10 identical cards, numbered 1, 2, 3,, 10, one card is drawn at random. Find the probability that the number on the card drawn is a multiple of :

(i) 2

(ii) 3

(iii) 2 and 3

(iv) 2 or 3

Answer

(i) Total number of possible outcomes = 10 (i.e., 1, 2, 3,, 10)

Number of favorable outcomes (Getting a card that is a multiple of 2) = 5 (2, 4, 6, 8, 10)

$$\begin{aligned} P(\text{Getting a card that is a multiple of 2}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{5}{10} \\ &= \frac{1}{2} \end{aligned}$$

Hence, the probability of getting a card that is a multiple of 2 is $\frac{1}{2}$.

(ii) Total number of possible outcomes = 10 (i.e., 1, 2, 3,, 10)

Number of favourable outcomes (Getting a card that is a multiple of 3) = 3 (3, 6, 9)

$$\begin{aligned} P(\text{Getting a card that is a multiple of 3}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{3}{10} \end{aligned}$$

Hence, the probability of getting a card that is a multiple of 3 is $\frac{3}{10}$.

(iii) Total number of possible outcomes = 10 (i.e., 1, 2, 3,, 10)

Number of favourable outcomes (Getting a card that is a multiple of 2 and 3) = 1 (6)

$P(\text{Getting a card that is a multiple of 2 and 3}) =$

$$\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{1}{10}$$

Hence, the probability of getting a card that is a multiple of 2 and 3 is $\frac{1}{10}$.

(iv) Total number of possible outcomes = 10 (i.e., 1, 2, 3,, 10)

Number of favourable outcomes (Getting a card that is a multiple of 2 or 3) = 7
(2, 3, 4, 6, 8, 9, 10)

$P(\text{Getting a card that is a multiple of 2 or 3}) =$

$$\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{7}{10}$$

Hence, the probability of getting a card that is a multiple of 2 or 3 is $\frac{7}{10}$.

Question 15

Two dice are thrown at the same time. Find the probability that the sum of the two numbers appearing on the top of the dice is :

(i) 0

(ii) 12

(iii) less than 12

(iv) less than or equal to 12

Answer

(i) Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),
.....(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (where the sum of the two numbers appearing
on the top of the dice is 0) = 0

$P(\text{where the sum of the two numbers appearing on the top of the dice is 0}) =$

$$\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{0}{36}$$

$$= 0$$

**Hence, the probability of the sum of the two numbers on the top of the
dice being 0 is 0.**

(ii) Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),
.....(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (where the sum of the two numbers appearing
on the top of the dice is 12) = 1 ((6, 6))

$$P(\text{where the sum of the two numbers appearing on the top of the dice is 12}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{1}{36}$$

Hence, the probability of the sum of the two numbers on the top of the dice being 12 is $\frac{1}{36}$.

(iii) Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (where the sum of the two numbers appearing on the top of the dice is less than 12) = 35 ((1, 1), (1, 2), (1, 3),(6, 5))

$$P(\text{where the sum of the two numbers appearing on the top of the dice is less than 12}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{35}{36}$$

Hence, the probability of the sum of the two numbers on the top of the dice being less than 12 is $\frac{35}{36}$.

(iv) Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (where the sum of the two numbers appearing on the top of the dice is less than or equal to 12) = 36 ((1, 1), (1, 2), (1, 3),(6, 5), (6, 6))

P(where the sum of the two numbers appearing on the top of the dice is less than or equal to 12) = $\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$

$$= \frac{36}{36}$$

$$= 1$$

Hence, the probability of the sum of the two numbers on the top of the dice being less than or equal to 12 is 1.

Question 16

A die is thrown once. Find the probability of getting :

- (i) a prime number
- (ii) a number greater than 3
- (iii) a number other than 3 and 5
- (iv) a number less than 6
- (v) a number greater than 6.

Answer

(i) Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a prime number) = 3 (2, 3, 5)

P(Getting the prime number) = $\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

Hence, the probability of getting the prime number is $\frac{1}{2}$.

(ii) Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a number greater than 3) = 3 (4, 5, 6)

$$P(\text{Getting a number greater than 3}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

Hence, the probability of getting a number greater than 3 is $\frac{1}{2}$.

(iii) Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a number other than 3 and 5) = 4 (1, 2, 4, 6)

$$P(\text{Getting a number other than 3 and 5}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

Hence, the probability of getting a number other than 3 and 5 is $\frac{2}{3}$.

(iv) Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a number less than 6) = 5 (1, 2, 3, 4, 5)

$$\begin{aligned} P(\text{Getting a number less than 6}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{5}{6} \end{aligned}$$

Hence, the probability of getting a number less than 6 is $\frac{5}{6}$.

(v) Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a number greater than 6) = 0

$$\begin{aligned} P(\text{Getting a number greater than 6}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{0}{6} \\ &= 0 \end{aligned}$$

Hence, the probability of getting a number greater than 6 is 0.

Question 17

Two coins are tossed together. Find the probability of getting :

(i) exactly one tail

(ii) at least one head

(iii) no head

(iv) at most one head

Answer

When two coins are tossed together, the total number of possible outcomes = 4
(i.e. HH, HT, TH and TT)

(i) Number of favourable outcomes (Getting exactly one tail) = 2 (HT and TH)

$$\begin{aligned}P(\text{Getting exactly one tail}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\&= \frac{2}{4} \\&= \frac{1}{2}\end{aligned}$$

Hence, the probability of getting exactly one tail is $\frac{1}{2}$.

(ii) Number of favourable outcomes (Getting at least one head) = 3 (HH, HT and TH)

$$\begin{aligned}P(\text{Getting at least one head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\&= \frac{3}{4}\end{aligned}$$

Hence, the probability of getting at least one head is $\frac{3}{4}$.

(iii) Number of favourable outcomes (Getting no head) = 1 (TT)

$$P(\text{Getting no head}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{1}{4}$$

Hence, the probability of getting no head is $\frac{1}{4}$.

(iv) Number of favourable outcomes (Getting at most one head) = 3 (HH, HT and TT)

$$\begin{aligned} P(\text{Getting at most one head}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{3}{4} \end{aligned}$$

Hence, the probability of getting at most one head is $\frac{3}{4}$.

Question 18

Two dice are thrown simultaneously, write all possible outcomes. Find :

- (i) probability of getting same number on both the dice.
- (ii) probability of getting a sum 7 on the uppermost faces of both the dice.

Answer

(i) Total number of possible outcomes = 36 (i.e., (1,1), (1,2), (1,3), (1,4), (1,5),.....(6,5), (6,6))

Number of favourable outcomes (Getting same number on both the dice) = 6
((1,1), (2,2), (3,3), (4,4), (5,5), (6,6))

$$\begin{aligned} P(\text{Getting same number on both the dice}) &= \\ &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \end{aligned}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6}$$

Hence, the probability of getting same number on both the dice is $\frac{1}{6}$.

(ii) Total number of possible outcomes = 36 (i.e., (1,1), (1,2), (1,3), (1,4), (1,5),.....(6,5), (6,6))

Number of favourable outcomes (Getting a sum 7 on the uppermost faces of both the dice) = 6 ((1,6), (2,5), (3,4), (4,3), (5,2), (6,1))

P(Getting a sum 7 on the uppermost faces of both the dice) =

$$\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6}$$

Hence, the probability of getting a sum 7 on the uppermost faces of both the dice is $\frac{1}{6}$.

Test Yourself

Question 1(i)

Two dice are rolled simultaneously. The probability of getting the sum equal to 5 is :

1. $\frac{1}{6}$

2. $\frac{2}{3}$

3. $\frac{1}{9}$

4. $\frac{4}{9}$

Answer

Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4), (1, 5),.....(6, 5), (6, 6))

Number of favourable outcomes (Getting the sum equal to 5) = 4 ((1, 4), (2, 3), (3, 2), (4, 1))

$$\begin{aligned}P(\text{Getting the sum equal to 5}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\&= \frac{4}{36} \\&= \frac{1}{9}\end{aligned}$$

Hence, option 3 is the correct option.

Question 1(ii)

A dice is rolled once. The probability of getting an odd number is :

1. $\frac{1}{6}$
2. 0.5
3. $\frac{1}{3}$
4. none of these

Answer

Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting an odd number) = 3 (1, 3, 5)

$$\begin{aligned}P(\text{Getting an odd number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\&= \frac{3}{6} \\&= \frac{1}{2}\end{aligned}$$

$$= 0.5$$

Hence, option 2 is the correct option.

Question 1(iii)

A card is drawn from a well shuffled pack of 52 playing cards. The probability of getting a club card is :

1. $\frac{3}{4}$

2. $\frac{2}{3}$

3. $\frac{1}{4}$

4. $\frac{1}{2}$

Answer

Total number of possible outcomes = 52

Number of favourable outcomes (Getting a club card) = 13

$$P(\text{Getting a club card}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{13}{52}$$

$$= \frac{1}{4}$$

Hence, option 3 is the correct option.

Question 1(iv)

A dice is thrown once. The probability of getting a number not more than 5 is :

1. $\frac{1}{2}$

2. $\frac{2}{3}$

3. $\frac{1}{6}$

4. $\frac{5}{6}$

Answer

Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a number not more than 5) = 4 (1, 2, 3, 4, 5)

$$\begin{aligned} P(\text{Getting a number not more than 5}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{5}{6} \end{aligned}$$

Hence, option 4 is the correct option.

Question 1(v)

A dice is rolled once. The probability of getting a prime number is :

1. $\frac{1}{2}$

2. $\frac{1}{3}$

3. $\frac{1}{4}$

4. $\frac{2}{3}$

Answer

Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting a prime number) = 3 (2, 3, 5)

$$P(\text{Getting a prime number}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{3}{6}$$

$$= \frac{1}{2}$$

Hence, option 1 is the correct option.

Question 1(vi)

Statement 1: Picking a red ball from a bag containing red balls is not a random experiment.

Statement 2: Random experiment is completely defined when we know all possible outcomes of the experiment but do not know which outcome will occur.

Which of the following options is correct?

1. Both the statements are true.

2. Both the statements are false.
3. Statement 1 is true, and statement 2 is false.
4. Statement 1 is false, and statement 2 is true.

Answer

A random experiment is defined as an experiment where there is uncertainty about the outcome, but the possible outcomes are known in advance.

So, statement 2 is true.

Picking a red ball from a bag containing only red balls is not a random experiment because there is only one possible outcome, which is getting a red ball.

So, statement 1 is true.

∴ Both the statements are true.

Hence, option 1 is the correct option.

Question 1(vii)

Assertion (A) : A dice is rolled two times the probability of getting an odd number on each dice is $\frac{1}{4}$.

Reason (R) : The favourable outcomes are (1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3) and (5, 6).

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.

3. A is true, but R is false.

4. A is false, but R is true.

Answer

Given,

A dice is rolled two times.

Possible outcomes = $\{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (3, 1), (3, 2), (3, 3), (3, 4), (3, 5), (3, 6), (4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (4, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$.

Number of total outcomes = 36

Favourable outcomes (for getting odd number on both dice) = $\{(1, 1), (1, 3), (1, 5), (3, 1), (3, 3), (3, 5), (5, 1), (5, 3), (5, 5)\}$

So, reason (R) is false.

Number of favourable outcomes = 9

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Number of total outcomes}} = \frac{9}{36} = \frac{1}{4}.$$

So, assertion (A) is true.

$\therefore$ A is true, but R is false.

Hence, option 3 is the correct option.

Question 1(viii)

Assertion (A) : Out of the given values: $\frac{1.5}{0.5}$, 1.1, 101% and -0.1, when asked which of them can be the probability of an event, a student answered -0.1.

Reason (R) : The probability of an event always lies between 0 and 1, both inclusive.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

The probability of an event always lies between 0 and 1, both inclusive.

∴ Reason (R) is true.

-0.1 cannot be the probability of an event as, probability cannot be negative.

∴ Assertion (A) is false.

Hence, option 4 is the correct option.

Question 1(ix)

Assertion (A) : When a dice is thrown the event of getting the first whole number is an impossible event.

Reason (R) : The probability of an event always lies between 0 and 1.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

A standard die has six faces, numbered 1 to 6.

The term "first whole number" typically refers to 0, which is indeed one of the impossible outcomes when a die is thrown.

Therefore, the event of getting the first whole number (i.e., 0) is a impossible event.

So, assertion (A) is true.

For any event A,

$$0 \leq P(A) \leq 1, \text{ not } 0 < P(A) < 1.$$

Thus, the probability of an event always lies between 0 and 1, including 0 and 1 as well.

So, reason (R) is false.

$\therefore$ A is true, but R is false.

Hence, option 3 is the correct option.

Question 1(x)

Assertion (A) : A bag contains red, white and blue pencils. The probability of selecting a red pencil is $\frac{2}{13}$ and that of selecting a blue pencil is $\frac{4}{13}$. Then the probability of selecting a white pencil will be $\frac{7}{13}$.

Reason (R) : The probability of all possible outcomes of an experiment must add upto 1.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are incorrect.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

The probability of selecting a red pencil = $\frac{2}{13}$

The probability of selecting a blue pencil = $\frac{4}{13}$

We know that,

The probability of all possible outcomes of an experiment must add upto 1.

So, reason (R) is true.

$$\Rightarrow P(\text{red}) + P(\text{blue}) + P(\text{white}) = 1$$

Let probability of selecting a white ball be x.

$$\Rightarrow \frac{2}{13} + \frac{4}{13} + x = 1$$

$$\Rightarrow \frac{6}{13} + x = 1$$

$$\Rightarrow x = 1 - \frac{6}{13}$$

$$\Rightarrow x = \frac{13}{13} - \frac{6}{13}$$

$$\Rightarrow x = \frac{13 - 6}{13}$$

$$\Rightarrow x = \frac{7}{13}.$$

So, assertion (A) is true.

$\therefore$ Both A and R are correct, and R is the correct explanation for A.

Hence, option 1 is the correct option.

Question 2

Two dice are rolled together. What is the probability of getting an odd number as sum ?

Answer

Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),
.....(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (Getting an odd number as sum) = 18 ((1, 2),
(1, 4), (1, 6), (2, 1), (2, 3), (2, 5),.....,(6, 3), (6, 5))

$$\begin{aligned}
 P(\text{Getting an odd number as sum}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\
 &= \frac{18}{36} \\
 &= \frac{1}{2}
 \end{aligned}$$

Hence, the probability of getting an odd number as the sum is $\frac{1}{2}$.

Question 3

Two dice are rolled together. What is the probability of getting a total of atleast 11 ?

Answer

Total number of possible outcomes = 36 (i.e., (1, 1), (1, 2), (1, 3), (1, 4),
.....(6, 4), (6, 5), (6, 6))

Number of favourable outcomes (Getting a total of at least 11) = 3 ((5, 6), (6, 5),
(6, 6))

$$\begin{aligned}
 P(\text{Getting a total of at least 11}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\
 &= \frac{3}{36} \\
 &= \frac{1}{12}
 \end{aligned}$$

Hence, the probability of getting a total of at least 11 is $\frac{1}{12}$.

Question 4

A card is drawn from a well shuffled deck of 52 cards. Find the probability of getting a black queen.

Answer

Total number of possible outcomes = 52

Number of favourable outcomes (Getting a black queen) = 2

$$\begin{aligned} P(\text{Getting a black queen}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{2}{52} \\ &= \frac{1}{26} \end{aligned}$$

Hence, the probability of getting a black queen is $\frac{1}{26}$.

Question 5

Find the probability that a leap year will have 53 Tuesdays.

Answer

In a leap year, there are 366 days.

366 days = 52 weeks + 2 days

These 2 days can be (Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun), and (Sun, Mon).

Total number of possible outcomes = 7

Number of favourable outcomes (Getting Tuesday as one of the extra days) = 2
(i.e., (Mon, Tue), (Tue, Wed)).

$$\begin{aligned} P(\text{Getting Tuesday as one of the extra days}) &= \\ &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{2}{7} \end{aligned}$$

Hence, the probability that a leap year will have 53 Tuesdays is $\frac{2}{7}$.

Question 6

Numbers 1 to 10 written on ten separate identical slips (one number on one slip) are kept in box and mixed well. One slip is chosen at random from the box without looking into it. What is the probability of :

- (i) getting a number less than 6 ?
- (ii) getting a single digit number ?

Answer

(i) Total number of possible outcomes = 10

Number of favourable outcomes (Getting a number less than 6) = 5 (1, 2, 3, 4, 5)

$$\begin{aligned} P(\text{Getting a number less than 6}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{5}{10} \end{aligned}$$

$$= \frac{1}{2}$$

Hence, the probability of getting a number less than 6 is $\frac{1}{2}$.

(ii) Total number of possible outcomes = 10

Number of favourable outcomes (Getting a single digit number) = 9 (1, 2, 3, 4, 5, 6, 7, 8, 9)

$$\begin{aligned} P(\text{Getting a single digit number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{9}{10} \end{aligned}$$

Hence, the probability of getting a single digit number is $\frac{9}{10}$.

Question 7

Find the probability of drawing a square number from a pack of 100 cards numbered from 1 to 100.

Answer

Total number of possible outcomes = 100

Number of favourable outcomes (Getting a square number) = 10 (1, 4, 9, 16, 25, 49, 64, 81, 100)

$$\begin{aligned} P(\text{Getting a square number}) &= \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{10}{100} \end{aligned}$$

$$= \frac{1}{10}$$

Hence, the probability of getting a square number is $\frac{1}{10}$.

Question 8

A dice is tossed once. What is the probability of the number 7, coming up ?

Answer

Total number of possible outcomes = 6 (i.e., 1, 2, 3, 4, 5, 6)

Number of favourable outcomes (Getting the number 7) = 0

$$P(\text{Getting the number 7}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{0}{6}$$

$$= 0$$

Hence, the probability of getting the number 7 is 0.

Question 9

A spinning wheel is divided into five equal sectors; out of which three are painted green, one is painted blue and the remaining one is painted red. What is the probability of getting a non-blue sector ?

Answer

Total number of possible outcomes = 5

Number of favourable outcomes (Getting a non-blue sector) = 4 (3 green + 1 red sector)

$$P(\text{Getting a non-blue sector}) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{4}{5}$$

Hence, the probability of getting a non-blue sector is $\frac{4}{5}$.

Question 10

A box contains 21 cards numbered 1, 2, 3, 4,, 21 and thoroughly mixed. A card is drawn at random from this box. What is the probability that the number on the card is divisible by 2 or 3 ?

Answer

Total number of possible outcomes = 21 (i.e., 1, 2, 3, 4,, 20, 21)

Number of favourable outcomes (Getting the number on the card that is divisible by 2 or 3) = 14 (2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21)

$P(\text{Getting the number on the card that is divisible by 2 or 3}) =$

$$\frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$= \frac{14}{21}$$

$$= \frac{2}{3}$$

Hence, the probability of getting the number on the card that is divisible by 2 or 3 is $\frac{2}{3}$.

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